K^R can detect complete intersection; so can conormal modules.

Def: Conormal module of a surjective ring hom. $\varphi: Q \twoheadrightarrow R$ with kernel I is: I/I^2.

Note I/I^2 is both Q & R-mod. (since I on I/I^2 is 0).

1. Lemmas.

① Free ideals.

Lem: R Noeth., I free mod. $\Rightarrow I = R$ or $I = (a)$ for a regular.

Pf:

- For any $x, y \in I$, $y \cdot x - x \cdot y = 0 \Rightarrow$ no L.I. elts $\Rightarrow I \cong R$.
- For any basis of I, $\quad R \xrightarrow{\pi} I$

$$r \longmapsto ra$$

is isom., $\ker(\pi) = \mathrm{ann}_R(a) = 0 \Rightarrow a$ regular or a is unit

② Dim. in fin. free exact seq.

Lem: $0 \to R^{\beta_d} \to \cdots \to R^{\beta_0} \to 0$ exact

$$\Rightarrow \sum_{i=0}^{d} (-1)^i \beta_i = 0.$$

③ Thm: $0 \neq I \trianglelefteq R$. $\mathrm{pdim}_R(I) < \infty \Rightarrow I$ contains a reg. elt.

Pf:

- Rank of I_p independent of P.

  Localising any finite free res. of I at P prime:

$$0 \to R_p^{\beta_d} \to \cdots \to R_p^{\beta_0} \to I_p \to 0$$

and if $P \in \mathrm{Ass}(R)$, R_P has no reg. elt.: any nonunit is $\frac{a}{b}$, $a \in P \Rightarrow a$ zero divisor.

Auslander-Buchsbaum $\Rightarrow \mathrm{pdim}_{R_p}(I_p) \le \mathrm{depth}(R_p) = 0 \Rightarrow I_p$ free.

by ② $\Rightarrow r := \mathrm{rank}(I_p) = \sum_{i=0}^{d} (-1)^i \mathrm{rank}(R_p^{\beta_i}) \Rightarrow$ independent of P.

- $\mathrm{rank}(I_p) \ge 1$.

Let $P \in \mathrm{Ass}_R(I)$. Note that any max. elt. of $\{\mathrm{ann}(x) : x \in I\}$ is ass. prime, so existence always holds.

$\begin{cases} P \in \mathrm{Ass}(I) \subset \mathrm{Ass}(R) \text{ by def.} \xrightarrow{\text{independence}} r \neq 0. \\ \mathrm{Ass}(I) \subset \mathrm{Supp}(I) \Rightarrow I_p \neq 0 \end{cases}$

- Contradiction.

I has no reg. elt. $\Rightarrow I \subset \bigcup_{P \in \mathrm{Ass}(R)} P \xrightarrow{\text{prime avoidance}} I \subset P$ for some $P \in \mathrm{Ass}(R)$.

Then $\begin{cases} I_p \neq R_p \text{ since othw. } \frac{i}{s_1} = \frac{1}{s_2} \Rightarrow i \notin P \\ I_p \neq 0 \Rightarrow I_p \text{ is free } R_p\text{-mod. of rank } r \ge 1. \end{cases}$

$\xrightarrow{①} I_p = (x)$ for $x \in R_p$ regular

$\Rightarrow \#$ since $P \in \mathrm{Ass}(R)$.

④ Tor_1 & reg. seq. on any M.

Lem:

i) $\underline{x}$ reg. seq. on $M \Rightarrow \mathrm{Tor}_1^R(M, R/(\underline{x}))=0$.

ii) $\underline{x}$ reg. seq. on M & $R \Rightarrow \mathrm{Tor}_i^R(M, R/(\underline{x}))=0 \ \forall\, i \geq 1$.

Pf: i) Consider the SES: let $I=(x_1,\cdots,x_n)$, $J=(x_1,\cdots,x_{n-1})$.

$$0 \to I/J \longrightarrow R/J \to R/I \to 0.$$

By this construction $I/J \hookrightarrow R/J$ ☆.

Then $\mathrm{Tor}_1^R(M, R/J) \to \mathrm{Tor}_1^R(M, R/I)$

$\mathrm{Tor}_1^R(M, R/J) = 0$ by induction

$$\hookrightarrow I \otimes M/JM \xrightarrow{\varphi} M/JM \longrightarrow M/IM \to 0.$$

But $\mathrm{im}\varphi$ is just $((x_n)+JM)/JM = \mathrm{im}(M/JM \xrightarrow{\cdot x_n} M/JM)$, if $\varphi(a_n \otimes m)$ $= a_n m = 0$ in $JM \Rightarrow m=0 \Rightarrow \varphi$ injective.

Thus $\mathrm{Tor}_1^R(M, R/I)=0$.

ii) Consider the SES: $I=(x_1,\cdots,x_n)$, $J=(x_1,\cdots,x_{n-1})$.

$$0 \to R/J \xrightarrow{\cdot x_n} R/J \longrightarrow R/I \to 0.$$

here $\cdot x_n$ injective since $\underline{x}$ reg. seq. on R.

Then by LES, $\mathrm{Tor}_i^R(M, R/I)=0 \ \forall\, i$. Note that $\mathrm{Tor}_1^R(M, R/I)=0$ since $M/JM \xrightarrow{\cdot x_n} M/JM$ injective.

Rmk: Let $R=k[[x,y]]/(xy)$, $M=k[[x,y]]/(x)$. Then M is R-mod. since $xy=0$ on M, but y is regular on M, not regular on R; $I:=(y)$.

Consider the res.:

$$\cdots \xrightarrow{\cdot x} R \xrightarrow{\cdot y} R \xrightarrow{\cdot x} \underset{\| \\ F_0}{R} \xrightarrow{R/(x)} M \longrightarrow 0$$

Tensoring $k[[x,y]]/(y)$:

$$\cdots \xrightarrow{\cdot x} R/(y) \xrightarrow{\cdot y} R/(y) \xrightarrow{\cdot x} R/(y) \longrightarrow 0$$

Then $\mathrm{Tor}_1^R(M, R/I) = 0$ since $\ker(\cdot x) = 0 = \mathrm{im}(\cdot y)$ (x reg. on $R/(y)$).

But $\mathrm{Tor}_2^R(M, R/I) \neq 0$ since $\ker(\cdot y) = R/(y) \neq \mathrm{im}(\cdot x) = (x)$

⑤ ☆ Thm: I gen. by reg. seq. $\Rightarrow I/I^2$ is free R/I-mod.

pf: By induction.

- Consider $R \xrightarrow{\pi} (a)/(a^2)$, $r \longmapsto ra$

  and $ra \in (a^2) \Rightarrow ra = ka^2$

  $\Rightarrow a(ka - r) = 0$

  a reg. $\Rightarrow r = ka \Rightarrow r \in I \Rightarrow R/I \cong (a)/(a^2)$.

- Let $J = (a_1, \cdots, a_{n-1})$, $I = (a_1, \cdots, a_n)$, consider

$$(R/I)^n \xrightarrow{\pi} I/I^2$$

$$(r_i)_i \longmapsto \sum r_i a_i.$$

$(r_i)_i \in \ker(\pi) \Rightarrow \sum r_i a_i = \sum s_i a_i$, $s_i \in I$

$$\Rightarrow (r_n - s_n) a_n \in J$$

(a) reg., $R/J \xrightarrow{\cdot a_n} R/J$

$$\Rightarrow r_n - s_n \in J$$

$$\Rightarrow r_n - s_n = \sum_{i=1}^{n-1} t_i a_i$$

$$\Rightarrow -\sum_{i=1}^{n-1} (r_i - s_i) a_i = \sum_{i=1}^{n-1} t_i a_i a_n$$

$$\Rightarrow \sum_{i=1}^{n-1} (t_i a_n + r_i - s_i) a_i = 0$$

$\Rightarrow t_i a_n + r_i - s_i \in I$ since $(R/I)^n \cong J/J^2$

$\xrightarrow{\text{tame}\in I} r_i - s_i \in I \xrightarrow{s_i \in I} r_i \in I.$

Thus π injective $\Rightarrow (R/\bar{I})^n \cong I/I^2$.

2. Vasconcelos' Thm.

① Thm: I is a nonzero ideal of finite proj. dim. of (Q, m, k).
$R = Q/I$. If I/I^2 is a free R-mod., then I is gen. by reg. seq.

Pf: Induction by existence of reg. elt. → Tor_1 to show finite proj. dim.

- ✡ Since I of fin. proj. dim., $\exists$ reg. elt. in I
  Thus $I \not\subset P \in \mathrm{Ass}(R)$, and $I \neq mI$

$\xrightarrow{\text{Prime Avoidance}} I \not\subset mI \cup \bigcup_{P \in \mathrm{Ass}(R)} P \Rightarrow \exists$ a minimal generator x of I that is reg. in R.

Then induction: $\begin{cases} n=1,\ I=(x) \text{ done} \\ \text{going to show } \mathrm{pdim}_{Q/(x)}(\bar{I}/(x)) < \infty \ \&\ (I/(x))/(\bar{I}/(x))^2 \end{cases}$

is free $R = (Q/(x))/(\bar{I}/(x))$ module.

Then I.H. is satisfied and $\bar{I}/(x)$ is gen. by reg. seq. over $Q/(x)$.

- Since x is reg. on R thus on I, Lemmas give:

$$\mathrm{Tor}_i^Q(Q/(x), I) = 0 \quad \forall\, i \geq 1,$$

and thus $H_i(F \otimes Q/(x)) = 0 \ \forall\, i \geq 1$, $H_0 = \mathrm{Tor}_0 = I/(x)I$, F res. of I,

therefore: $F \otimes Q/(x)$ resolves $I/(x)I$ $\xrightarrow{I \text{ has fin. proj. dim.}}$ $\mathrm{pdim}_{Q/(x)}(I/(x)I) < \infty$.

- Remains to show $I/(x)$ is a direct summand of $I/(x)I$.

i) extends x to a basis of I/I^2 (not I !! since we need info on I).

This can be done since $(I/I^2)/m(I/I^2) \cong I/mI$

Let $J=(x_1,\cdots,x_{n-1})$ be such that $I=J+(x)$.

ii) $I/(x)=(J+(x))/(x)=J/J\cap(x) \hookrightarrow I/J\cap(x)$

Observe $J\cap(x)$:

$y\in J\cap(x) \Rightarrow y=ax=\sum_{i=1}^{n-1} a_i x_i \Rightarrow ax-\sum_{i=1}^{n-1} a_i x_i=0$

$\xrightarrow{\text{l.I. in } I/I^2} a\in I^2\subset I.$

$\Rightarrow y\in(x)I \Rightarrow J\cap(x)\subset(x)I\subset(x)\cdot R$

Thus $I/J\cap(x) \twoheadrightarrow I/(x)I \twoheadrightarrow I/(x)$.

Thus $I/(x) \longrightarrow I/(x)I \longrightarrow I/(x)$

$x_i \longmapsto x_i \longmapsto x_i$

$\Rightarrow$ composition is id.

Thus $I/(x)$ direct summand of $I/(x)I$.

$\Rightarrow \operatorname{pdim}_{R/(x)}(I/(x))<\infty$.

- $I/(x)/(I/(x))^2$ is free.

$\begin{cases} (I/(x))^2=(I^2+(x))/(x) \text{ by definition, and} \\ I/(x)/(I/(x))^2 = I/(I^2+(x)) = (I/I^2)/(x+I^2) \text{ is free.} \end{cases}$

free by hypothesis; image of x in I/I^2, simple mod.

This completes the induction.

② Coro.:

R is C.I. $\Leftrightarrow \exists$ reg. repn. $\hat{R}\cong Q/I$ s.t. I/I^2 free $\hat{R}$-mod.

$\Leftrightarrow \forall$ reg. repn. $\hat{R}\cong Q/I$, I/I^2 is free $\hat{R}$-mod.

Pf: "⇒": I gen. by reg. seq. ⇒ I/I^2 is free Q/I-mod.

"⇐": Note that by A-B-S, since Q is regular, I f.g. has finite proj. dim.

Thus I/I^2 free ⇒ I gen. by reg. seq. ⇒ R C.I.